CIS-445 Machine Learning
Group Homework #4

Names of Group
Members & Group #: **Group #5 Dylan Shea, Clay Peterson, Channel Caldwell**
Logistics

1. Get in touch with your group. (See Groups folder on Blackboard.)
2. Discuss and work all 3 problems collectively with your group via E-mail, Discussion Forum, Blackboard Collaborate Ultra, and/or MS Teams. (Do not divide the work among group members.)
3. Choose a recorder to prepare the final copy (one per group) and submit it via the Blackboard Assignments/Homeworks folder.
4. Be sure all group members' names are on final copy. Do **not** add names of your group members who did not participate in the assignment or whose contribution was minimal.

Worth 50 points. Due date: See Blackboard

Objectives: Learn about Statistical Methods: Descriptive Statistics, Linear Regression Analysis, and Logistic Regression Analysis

Work the following Problems and provide answers on these sheets.
You must show your calculations!

1. Ex. 2 (a through f), p. 194, Kantardzic
   n=20

a) 12 + 5 + 18 + 5 + 9 + 13 + 12 + 19 + 7 + 12 + 12 + 13 + 3 + 4 + 5 + 13 + 8 + 7 + 6 = 190
190/20= 9.5 is the mean

b) First sort the data, 3, 4, 5, 5, 5, 6, 7, 7, 7, 8, 9, 12, 12, 12, 12, 13, 13, 13, 18, 19
Then find the one in the middle which are two (8 and 9)
9+8=17/2=8.5 is the median

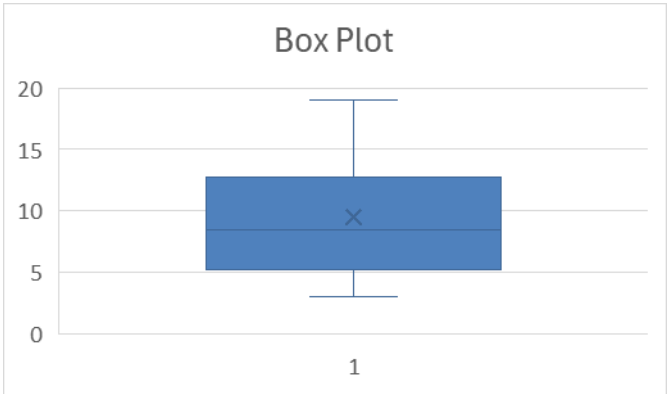
c) Mode is 12 because it appears four times. Modality is unimodal.
d) Standard Deviation: 4.5364

Xi-MEAN of X	(Xi-MEAN)^2
7-9.5=-2.5	6.25
12-9.5=2.5	6.25
5-9.5=-4.5	20.25
18-9.5=8.5	72.25
5-9.5=-4.5	20.25
9-9.5=-.5	0.25
13-9.5=3.5	12.25
12-9.5=2.5	6.25
19-9.5=9.5	90.25
7-9.5=-2.5	6.25
12-9.5=2.5	6.25
12-9.5=2.5	6.25

13-9.5=3.5	12.25
3-9.5=-6.5	42.25
4-9.5=-5.5	30.25
5-9.5=-4.5	20.25
13-9.5=3.5	12.25
8-9.5=-1.5	2.25
7-9.5=-2.5	6.25
6-9.5=-3.5	12.25
SUM:	391
Variance:	391/(20-1) = 20.5789
Standard Deviation:	(20.5789)^1/2=4.5364

e) Box plot:
First sort the data, 3, 4, 5, 5, 5, 6, 7, 7, 7, 8, 9, 12, 12, 12, 12, 13, 13, 13, 18, 19

Min=3, Max =19
Mean= 8.5
Q1= (n+1)/4=21/4=5.25
Q3=3*21/4=15.75 (between 12 and 13)= 12.75
IQR= 12.75-5.25=7.5



f) To find the outliers: Lower: mean – 3 * Standard D = 9.5-3*4.54 = -4.12
Upper: mean + 3 * Standard D = 9.5-3*4.54 = 23.12
There are no outliers since none of the data is < -4.12 or > 23.12

2. Ex. 4 (a and b only), p. 195, Kantardzic

a.

	A.			
X	1	4	3	5
	13	3.25		
Y	5	2.75	3	2.5
	13.25	3.3125		

Slope

X	Y	x - x mean	y - y mean	product	x - x mean ^2
1	5	-2.25	1.6875	-3.79688	5.0625
4	2.75	0.75	-0.5625	-0.42188	0.5625
3	3	-0.25	-0.3125	0.078125	0.0625
5	2.5	1.75	-0.8125	-1.42188	3.0625

Sum of X – mean and Y-mean Add em up

-5.5625

Add Em up

8.75

divide these 2 to get slope

-0.63571

Y-Intercept

5.378571

b.

b.	y-y mean ^2
	2.85
	0.32
	0.10
	0.66
sum	3.92
r	-0.950

3. Ex. 5 (a, b, c), p. 195, Kantardzic

5. A logit function, obtained through logistic regression, has the form

$$\text{Logit}(p) = 1.2 - 1.3 \times 1 + 0.6 \times 2 + 0.4 \times 3$$

Find the probability of output values 0 and 1 for the following samples.

(a) {1, −1, −1}
(b) {−1, 1, 0 }
(c) {0, 0, 0 }

- A

x1	x2	x3	
1	**-1**	**-1**	
logit(p)=	**-1.1**		=1.2-1.3*A2+0.6*B2+0.4*C2
p(Y=1)	**0.25**		=EXP(B4)/(1+EXP(B4))
p(Y=0)	**0.75**		=1-B6

- B

x1	x2	x3	
-1	**1**	**0**	
logit(p)=	**3.1**		=1.2-1.3*A2+0.6*B2+0.4*C2
p(Y=1)	**0.96**		=EXP(B4)/(1+EXP(B4))
p(Y=0)	**0.04**		=1-B6

- C

x1	x2	x3	
0	**0**	**0**	
logit(p)=	**1.2**		=1.2-1.3*A2+0.6*B2+0.4*C2
p(Y=1)	**0.77**		=EXP(B4)/(1+EXP(B4))
p(Y=0)	**0.23**		=1-B6